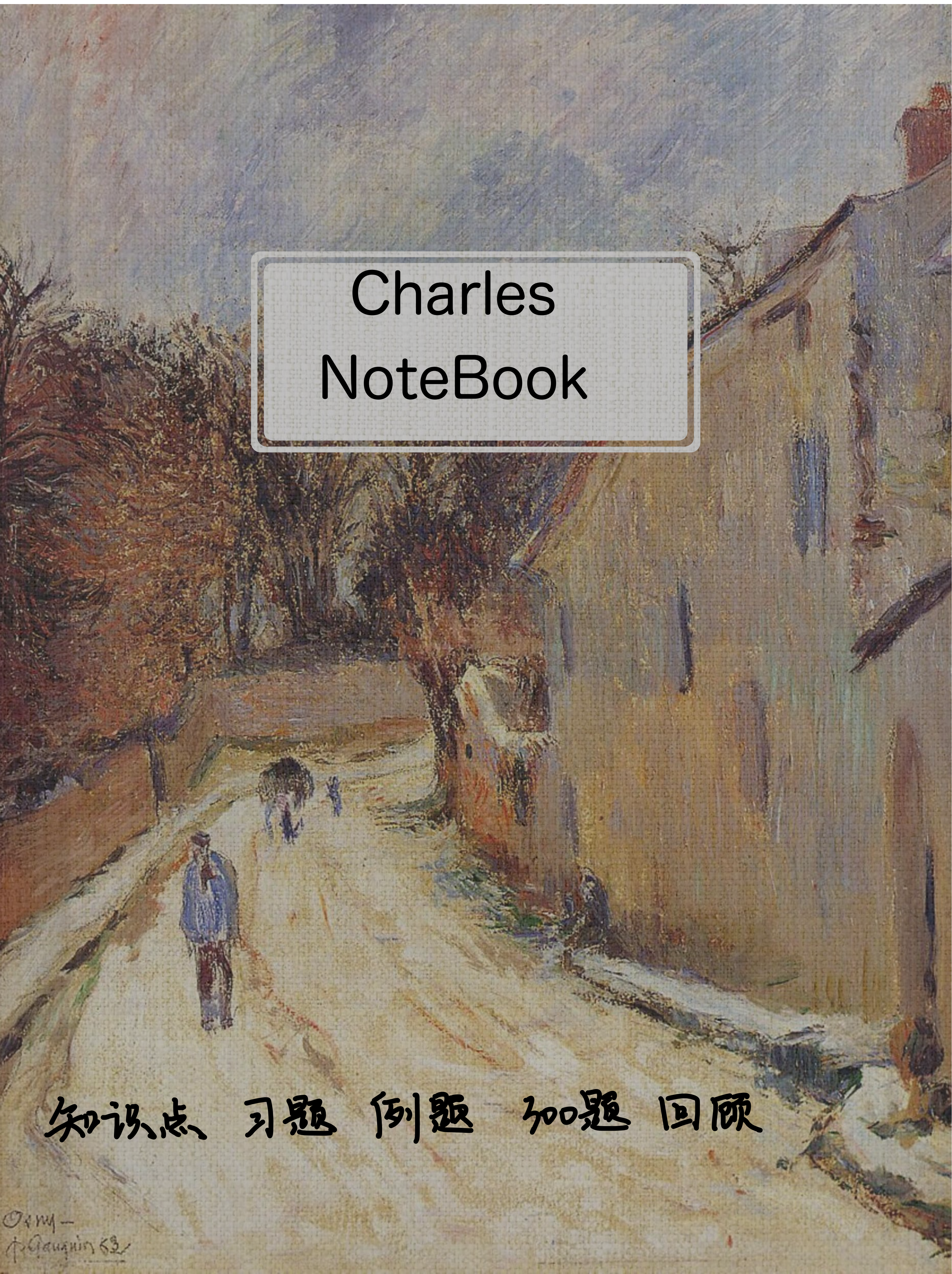


Charles NoteBook

知识点、习题 例题 加题 回顾

二重积分

性质：①区域面积 $\iint_D 1\,d\sigma = A$　②f(x,y) D上有界　③线性　④可加性（可拆）

⑤保号性：$\begin{cases} f(x,y)\le g(x,y) \Rightarrow \iint_D f\,d\sigma \le \iint_D g\,d\sigma \\ \left|\iint_D f\,d\sigma\right| \le \iint_D |f|\,d\sigma \end{cases}$

⑥估值定理　$mA \le \iint_D f(x,y)\,d\sigma \le MA$

★⑦中值定理　$\iint_D f(x,y)\,d\sigma = f(\xi,\eta)\cdot A$　（新增知识点）

[注（2023版1000题）

设 $f(x,y)$ 连续，$f(0,0)$ 已知，$D=\{(x,y)\mid x^2+y^2\le t^2\}$

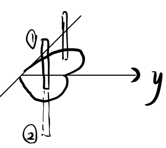
则 $\lim_{t\to0^+}\frac{1}{\pi t^2}\iint_D f(x,y)\,d\sigma = ?$

原式 $= \lim_{t\to0^+}\frac{1}{\pi t^2}\cdot f(\xi,\eta)\cdot A = \lim_{t\to0^+}\frac{f(\xi,\eta)\cdot\pi t^2}{\pi t^2} = \lim_{(\xi,\eta)\to(0,0)} f(\xi,\eta) \overset{\text{连续}}{=} f(0,0)$

普通对称性与轮换对称性

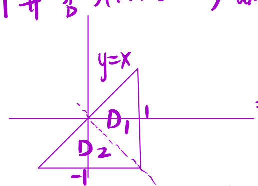
①普通对称性

如D关于y轴对称：$\iint_D f(x,y)\,d\sigma = \begin{cases} 2\iint_{D_1} f(x,y)\,d\sigma, & f(x,y)=f(-x,y) \\ 0, & f(x,y)=-f(-x,y) \end{cases}$



例12.2的一部分

计算 $\iint_D y\left(1+xe^{\frac{x^2+y^2}{2}}\right)dxdy$，其中区域D由直线 $y=x$，$y=-1$ 及 $x=1$ 所围成.



$\iint_D y + xy\frac{x^2+y^2}{2}\,d\sigma$

$= \underbrace{\iint_D y\,d\sigma}_{} + \underbrace{\iint_D xy\frac{x^2+y^2}{2}\,d\sigma}_{}$

D_1上 $f(x,y)= xy\frac{x^2+y^2}{2} = -\left(-xy\frac{x^2+y^2}{2}\right) = -f(x,-y)$

D_1上 $f(x,y) = y = -f(x,-y)$　　D_2上 $f(x,y) = xy\frac{x^2+y^2}{2} = -f(-x,y)$

D_2上 $f(x,y) = y = f(-x,y)$　　原 $= 2\iint_{D_2} y\,d\sigma$

②轮换对称性（直角坐标系）

$\iint_{D_1:\frac{x^2}{4}+\frac{y^2}{3}\le1}(2x^2+3y^2)\,dxdy \overset{?}{=} \iint_{D_2:\frac{y^2}{4}+\frac{x^2}{3}\le1}(2y^2+3x^2)\,dydx$

$\iint_{D:\frac{x^2}{4}+\frac{y^2}{4}\le1}(2x^2+3y^2)\,dxdy = \iint_{D:\frac{x^2}{4}+\frac{y^2}{4}\le1}(2y^2+3x^2)\,dydx$　（D关于 $y=x$ 对称）

$x\leftrightarrow y$

字母对调后，区域D不变，则称轮换对称性

（eg　若 $f(x,y)+f(y,x)=a$）

$\Rightarrow I=\frac{1}{2}\iint_D [f(x,y)+f(y,x)]\,d\sigma = \frac{a}{2}A$

注例

设区域 $D=\{(x,y)\mid x^2+y^2\le 1,\ x\ge 0,\ y\ge 0\}$，$f(x)$为D上正值连续函数，$a,b$常数

求 $I=\iint\limits_D \dfrac{a\sqrt{f(x)}+b\sqrt{f(y)}}{\sqrt{f(x)}+\sqrt{f(y)}}\,d\sigma$

$$I=\iint\limits_D \frac{a\sqrt{f(x)}+b\sqrt{f(y)}}{\sqrt{f(x)}+\sqrt{f(y)}}\,d\sigma=\iint\limits_D \frac{a\sqrt{f(y)}+b\sqrt{f(x)}}{\sqrt{f(x)}+\sqrt{f(y)}}\,d\sigma$$

$$I\Rightarrow \frac{a+b}{2}\iint\limits_D \frac{\sqrt{f(x)}+\sqrt{f(y)}}{\sqrt{f(x)}+\sqrt{f(y)}}\,d\sigma=\frac{a+b}{2}\cdot A=\frac{a+b}{2}\cdot\frac{\pi}{4}$$

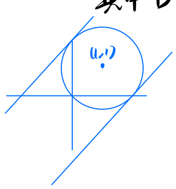
例12.1

设 $I_1=\iint\limits_D \sin\left|\dfrac{x-y}{2}\right|dxdy$，

$I_2=\iint\limits_D \sin\left(\dfrac{x-y}{2}\right)^2 dxdy$

$I_3=\iint\limits_D \sin\left(\dfrac{x-y}{2}\right)^3 dxdy$

其中 $D=\{(x,y)\mid (x-1)^2+(y-1)^2\le 2\}$



$\dfrac{x-y}{2}\to\ y=x+b$

$y=x+2$ 与 $y=x-2$ 之间

$x-2<y<x+2$

$-1<\dfrac{x-y}{2}<1$

$0<\left|\dfrac{x-y}{2}\right|<1$

$\sin x$ 在 $(0,1)$ ↗

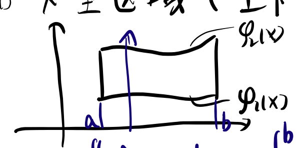
$\sin\left|\dfrac{x-y}{2}\right| > \sin\left(\dfrac{x-y}{2}\right)^2>0$

$\Rightarrow$ 保号性：$I_1>I_2>0$

I_3：$I_3=\dfrac{1}{2}\iint\limits_D \sin\dfrac{(x-y)^3}{2}+\sin\dfrac{(y-x)^3}{2}\,dx=0.\quad\Rightarrow\quad I_1>I_2>I_3$

二 计算

(1) X型区域（上下型）

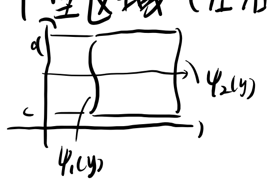


后积先定限，限内画条线

先交写下限，后交写上限.

$$\iint\limits_D f(x,y)\,d\sigma=\int_a^b dx\int_{\varphi_1(x)}^{\varphi_2(x)} f\,dy$$

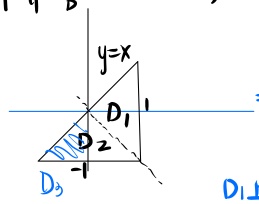
(2) Y型区域（左右型）



$$\int_c^d dy\int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\,dx$$

例12.2

计算 $\iint_D y(1+xe^{\frac{x^2+y^2}{2}})\,dxdy$，其中区域D由直线 $y=x$, $y=-1$ 及 $x=1$ 所围成.



$$\iint_D y + xy\frac{x^2+y^2}{2}\,d\sigma$$

$$= \iint_D y\,d\sigma + \iint_D xy\frac{x^2+y^2}{2}\,d\sigma$$

D_1上 $f(x,y)=xy\frac{x^2+y^2}{2} = -(-xy\frac{x^2+y^2}{2}) = -f(x,-y)$

D_1上 $f(x,y)= y=-f(x,-y)$ D_2上 $f(x,y)=xy\frac{x^2+y^2}{2} = -f(-x,y)$

D_2上 $f(x,y)= y= f(-x,y)$ 原 $= 2\iint_{D_3} y\,d\sigma = 2\int_{-1}^{0}dx\int_{-1}^{x} y\,dy$

$$= \int_{-1}^{0} x^2-1\,dx = -\frac{2}{3}$$

2. 极坐标系下 $d\theta\,dr$ $dr\,d\theta$

$$\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases} \qquad \iint_D f(x,y)\,d\sigma = \int_\alpha^\beta d\theta\int_{r_1(\theta)}^{r_2(\theta)} f\,r\,dr$$

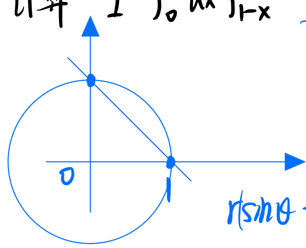
3. 极直选择原则:

☆ 1). 被积函数是形为 $f(x^2+y^2)$, $f(\frac{y}{x})$, $f(\frac{x}{y})$ 全要看!

2) 积分区域是圆（的一部分）

例12.6

计算 $I=\int_0^1 dx\int_{1-x}^{\sqrt{1-x^2}} \frac{x+y}{x^2+y^2}\,dy$



$y=1-x$

$y=\sqrt{1-x^2}$

$r(\sin\theta+\cos\theta)=1$

用极坐标系

$$I=\int_0^{\frac{\pi}{2}} d\theta\int_{\frac{1}{\sin\theta+\cos\theta}}^{1} \frac{r(\sin\theta+\cos\theta)}{r^2}\,r\,dr$$

$$I=\int_0^{\frac{\pi}{2}} d\theta\int_{\frac{1}{\sin\theta+\cos\theta}}^{1} (\sin\theta+\cos\theta)\,dr$$

$$I=\int_0^{\frac{\pi}{2}} (\sin\theta+\cos\theta)(1-\frac{1}{\sin\theta+\cos\theta})\,d\theta$$

$$I=\int_0^{\frac{\pi}{2}} \sin\theta+\cos\theta-1\,d\theta$$

$$I=-\cos\theta+\sin\theta-\theta\Big|_0^{\frac{\pi}{2}} = (0+1-\frac{\pi}{2})-(-1+0-0)=2-\frac{\pi}{2}$$

4. 积分次序

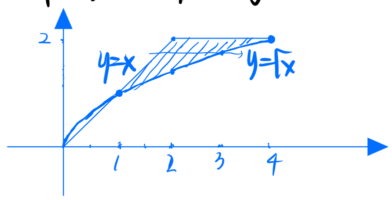
☆☆ 有原函数但不能初等函数表达的:

$\int\sin\frac{1}{x}dx$, $\int\cos\frac{1}{x}dx$, $\int\frac{\sin x}{x}dx$, $\int\frac{\cos x}{x}dx$, $\int\frac{\tan x}{x}dx$, $\int\frac{e^x}{x}dx$, $\int\sin x^2dx$

$\int\cos x^2dx$, $\int\tan x^2dx$, $\int e^{ax^2+bx+c}dx$ $(a\neq0)$ (如 $\int e^{x^2}dx$, $\int e^{-x^2}dx$, $\int\frac{dx}{\ln x}$)

例12.7

计算 $\int_1^2 dx\int_{\sqrt{x}}^{x}\sin\frac{\pi x}{2y}dy+\int_2^4 dx\int_{\sqrt{x}}^{2}\sin\frac{\pi x}{2y}dy$



原 $=\int_1^2 dy\int_y^{y^2}\sin\frac{\pi x}{2y}dx$

$=\int_1^2 -\frac{2y}{\pi}\cos\frac{\pi}{2y}x\Big|_y^{y^2}dy$ $\qquad$ $y \quad 1 \quad 0$; $\cos\frac{\pi}{2}y \quad \frac{2}{\pi}\sin\frac{\pi}{2}y \quad \frac{-4}{\pi^2}\cos\frac{\pi}{2}y$

$=\int_1^2 \frac{-2y}{\pi}\left(\cos\frac{\pi}{2}y\right)dy=\frac{-2}{\pi}\int_1^2 y\cos\frac{\pi}{2}y\,dy$

$=-\frac{2}{\pi}\left[\frac{2y}{\pi}\sin\frac{\pi}{2}y\Big|_1^2+\frac{4}{\pi^2}\cos\frac{\pi}{2}y\Big|_1^2\right]$

$=-\frac{2}{\pi}\left[-\frac{2}{\pi}-\frac{4}{\pi^2}\right]=\frac{4}{\pi^2}+\frac{8}{\pi^3}$

5.用二重积分处理一元积分问题

例12.8

设 $f(x)=\int_x^1\sin(\pi u^2)du$,求 $\int_0^1 f(x)dx$

$\int_0^1 dx\int_x^1\sin(\pi u^2)du$

$=\int_0^1 du\int_0^u\sin(\pi u^2)dx=\int_0^1 u\sin(\pi u^2)du=\frac{1}{2\pi}\int_0^1\sin(\pi u^2)d\pi u^2$

$=\frac{-1}{2\pi}\cos(\pi u^2)\Big|_0^1=\frac{-1}{2\pi}(-1-1)=\frac{1}{\pi}$

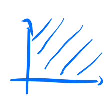
例12.10 ★★★

$\int_0^{+\infty}e^{-x^2}dx$

$I=\int_0^{+\infty}e^{-x^2}dx$

$I^2=\int_0^{+\infty}e^{-x^2}dx\cdot\int_0^{+\infty}e^{-x^2}dx$

$=\int_0^{+\infty}e^{-x^2}dx\cdot\int_0^{+\infty}e^{-y^2}dy$ ★

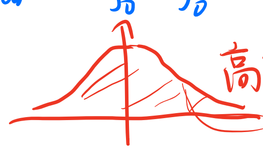


$=\iint_D e^{-(x^2+y^2)}d\sigma$

$=\int_0^{\frac{\pi}{2}}d\theta\int_0^{+\infty}e^{-r^2}rdr=\frac{1}{2}\int_0^{\frac{\pi}{2}}d\theta\int_0^{+\infty}e^{-r^2}dr^2=\frac{\pi}{4}$

$I=\frac{\sqrt{\pi}}{2}$

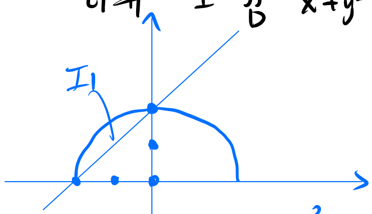
$\Rightarrow$



(2022)

已知平面区域 $D=\{(x,y)\mid y-2\le x\le\sqrt{4-y^2},\ 0\le y\le 2\}$

计算 $I=\iint_D \frac{(x-y)^2}{x^2+y^2}\,dxdy$



$y=x+2$

$x^2+y^2=4$

$r(\sin\theta-\cos\theta)=2$

$$I_1=\int_{\frac{\pi}{2}}^{\pi}d\theta\int_{\frac{2}{\sin\theta-\cos\theta}}^{2}\frac{r^2(\cos\theta-\sin\theta)^2}{r^2}\cdot r\,dr$$

$$=\int_{\frac{\pi}{2}}^{\pi}(1-\sin2\theta)\,d\theta\int_{\frac{2}{\sin\theta-\cos\theta}}^{2}r\,dr$$

$$=\int_{\frac{\pi}{2}}^{\pi}\left(2-\frac{2}{1-\sin2\theta}\right)(1-\sin2\theta)\,d\theta$$

$$=\int_{\pi/2}^{\pi}2-2\sin2\theta-2\,d\theta=\cos2\theta\Big|_{\pi/2}^{\pi}=2$$

$$I_2=\int_0^{\pi}d\theta\int_0^2\frac{r^2(1-\sin2\theta)}{r^2}\,r\,dr$$

$$=\int_0^{\pi}(1-\sin2\theta)\,d\theta\int_0^2 r\,dr$$

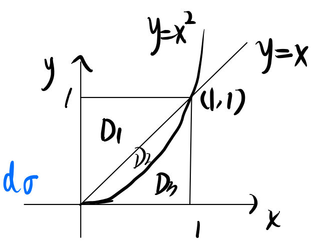
$$=2\int_0^{\pi}1-\sin2\theta\,d\theta=2\left[\theta+\frac{1}{2}\cos2\theta\right]_0^{\pi}=2\left[\pi+\frac{1}{2}-0-\frac{1}{2}\right]=2\pi$$

$$\Rightarrow I=I_2-I_1=2\pi-2$$

例12.3

计算 $\iint_D |y-x^2|\cdot\max\{x,y\}\,dxdy$

其中 $D=\{(x,y)\mid 0\le x\le1,\ 0\le y\le1\}$



$$\iint_{D_1}(y-x^2)\cdot y\,d\sigma+\iint_{D_2}(y-x^2)x\,d\sigma+\iint_{D_3}(x^2-y)x\,d\sigma$$

$$=\int_0^1dx\left[\int_x^1(y-x^2)y\,dy+\int_{x^2}^{x}(y-x^2)x\,dy+\int_0^{x^2}(x^2-y)x\,dy\right]$$

$$=\int_0^1\frac{y^3}{3}-\frac{x^2y^2}{2}\Big|_{y=x}^{y=1}+\frac{xy^2}{2}-x^3y\Big|_{y=x^2}^{y=x}+x^3y-\frac{xy^2}{2}\Big|_{y=0}^{y=x^2}dx$$

$$=\int_0^1\left(\frac{1}{3}-\frac{x^2}{2}\right)-\left(\frac{x^3}{3}-\frac{x^4}{2}\right)+\left(\frac{x^3}{2}-x^4\right)-\left(\frac{x^5}{2}-x^5\right)+\left(x^5-\frac{x^5}{2}\right)dx$$

$$=\int_0^1\frac{1}{3}-\frac{x^2}{2}+\frac{x^3}{6}-\frac{1}{2}x^4+x^5\,dx$$

$$=\frac{1}{3}x-\frac{x^3}{6}+\frac{x^4}{24}-\frac{x^5}{10}+\frac{x^6}{6}\Big|_0^1=\frac{1}{3}+\frac{1}{24}-\frac{1}{10}=\frac{11}{40}$$

例12.4

设区域 $D=\{(x,y)\mid x^2+y^2\le R^2\}$ 计算 $\iint_D\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)dxdy$

$$\iint_D\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)d\sigma=\iint_D\frac{y^2}{a^2}+\frac{x^2}{b^2}\,d\sigma$$

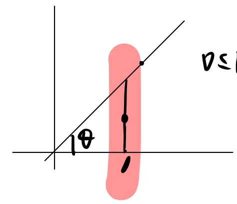
$$\Rightarrow I=\frac{1}{2}\iint_D\frac{x^2+y^2}{a}+\frac{x^2+y^2}{b^2}\,d\sigma=\frac{1}{2}\int_0^{2\pi}d\theta\int_0^R\left(\frac{1}{a^2}+\frac{1}{b^2}\right)r^3\,dr$$

$$=\frac{1}{2}\left(\frac{1}{a^2}+\frac{1}{b^2}\right)\int_0^{2\pi}\frac{R^4}{4}\,d\theta=\frac{R^4\pi}{4}\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$$

例12.5

计算二重积分 $I=\iint\limits_D r^2\sin\theta\sqrt{1-r^2\cos2\theta}\,dr\,d\theta$

其中 $D=\left\{(r,\theta)\,\middle|\,0\le r\le\sec\theta,\ 0\le\theta\le\frac{\pi}{4}\right\}$



$0\le r\le\frac{1}{\cos\theta}$

$$I=\iint(r\sin\theta)\sqrt{1-(r^2\cos^2\theta-r^2\sin^2\theta)}\;r\,dr\,d\theta$$

$$=\iint\limits_D y\sqrt{1-x^2+y^2}\,d\sigma$$

$$=\int_0^1dx\int_0^x y\sqrt{1-x^2+y^2}\,dy$$

$$=\frac12\int_0^1dx\int_0^x\sqrt{(1-x^2)+y^2}\,dy^2$$

$\sqrt{a+x}\,dx$

$\frac23(a+x)^{\frac32}$

$$=\frac13\int_0^1(1-x^2+y^2)^{\frac32}\Big|_{y=0}^{y=x}dx$$

$$=\frac13\int_0^1 1-(1-x^2)^{\frac32}dx\overset{x=\sin t}{=}\frac23-\frac23\int_0^{\frac{\pi}{2}}\cos^3t\cdot\cos t\,dt$$

$$=\frac13-\frac13\cdot\frac34\,\frac12\,\frac{\pi}{2}=\frac13-\frac{\pi}{16}$$

例12.9

设 $f(x)$ 为恒大于零的连续函数，证明 $\int_0^1f(x)dx\cdot\int_0^1\frac{1}{f(x)}dx\ge1$

$$\int_0^1f(x)\,dx\cdot\int_0^1\frac{1}{f(y)}\,dy$$

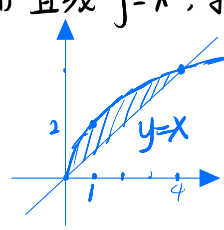
$$\int_0^1\int_0^1\frac{f(x)}{f(y)}\,dx\,dy=\iint\frac{f(x)}{f(y)}\,d\sigma=\iint\frac{f(y)}{f(x)}\,d\sigma$$

$$=\frac12\iint\frac{f(x)}{f(y)}+\frac{f(y)}{f(x)}\,d\sigma\ge\frac12\iint2\sqrt{\frac{f(x)}{f(y)}\cdot\frac{f(y)}{f(x)}}\,d\sigma=\iint1\,d\sigma=1$$

习12.1

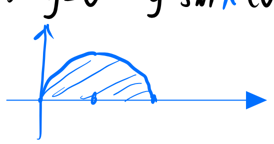
按两种不同积分次序化二重积分 $\iint\limits_D f(x,y)\,dx\,dy$ 为累次积分，其中 D 为：

(1) 直线 $y=x$，抛物线 $y^2=4x$ 所围闭区域.



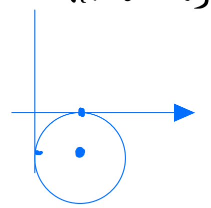
$$\int_0^4dx\int_x^{2\sqrt{x}}f(x,y)\,dy$$

(2) $y=0$ $\quad y=\sin x\ (0\le x\le\pi)$



$$\int_0^{\pi}dx\int_0^{\sin x}f(x,y)\,dy$$

(3) $(x-1)^2+(y+1)^2 \le 1$

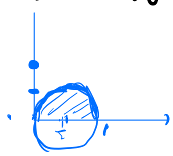


$\sqrt{1-(x-1)^2}$

$$\int_0^2 dx \int_{-\sqrt{1-(x-1)^2}-1}^{\sqrt{1-(x-1)^2}-1} f(x,y)\,dy$$

习12.2

$$\int_0^{\frac{\pi}{2}} d\theta \int_0^{\cos\theta} f(r\cos\theta, r\sin\theta)\, r\,dr$$ 可写成



$r=\cos\theta$ $\quad y^2+x^2-x+\frac{1}{4}=\frac{1}{4}$ $\quad y^2+(x-\frac{1}{2})^2=\frac{1}{4}$

$r^2=x=x^2+y^2 \Rightarrow y^2=x-x^2$

$I=\int_0^1 dx\int_0^{\sqrt{x-x^2}} f(x,y) \Rightarrow D$

习12.3

证明: $\iint_D (e^{\sin y}+e^{-\sin x})\,dxdy \ge 2\pi^2$, 其中 $D=\{(x,y)\,|\,0\le x\le\pi,\ 0\le y\le\pi\}$

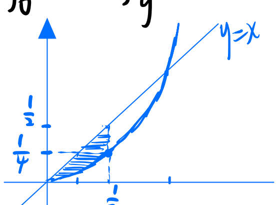
$I=\frac{1}{2}\iint_D (e^{\sin y}+e^{\sin x}+e^{-\sin x}+e^{-\sin y})\,d\sigma$

$\ge \frac{1}{2}\iint 2\sqrt{e^{\sin x}+e^{-\sin x}}+2\sqrt{e^{-\sin y}+e^{\sin y}}\,d\sigma$

$=2\iint d\sigma = 2\pi^2$

习12.4

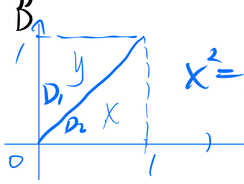
$$\int_0^{\frac{1}{4}} dy\int_y^{\sqrt{y}} f(x,y)\,dx + \int_{\frac{1}{4}}^{\frac{1}{2}} dy\int_y^{\frac{1}{2}} f(x,y)\,dx$$



原 $=\int_0^{\frac{1}{2}} dx\int_{x^2}^{x} f(x,y)\,dy$

习12.5

$\iint_D e^{\max\{x^2,y^2\}}\,dxdy$, 其中 $D=\{(x,y)\,|\,0\le x\le 1,\ 0\le y\le 1\}$



$x^2=y^2$ $\quad \iint_{D_1} e^{y^2}d\sigma+\iint_{D_2} e^{x^2}d\sigma=\int_0^1 e^{y^2}dy\int_0^y dx+\int_0^1 e^{x^2}dx\int_0^x dy$

$=\frac{1}{2}\int_0^1 e^{y^2}dy^2+\frac{1}{2}\int_0^1 e^{x^2}dx^2=\int_0^1 e^{x^2}dx^2=e^{x^2}\Big|_0^1=e-1$

习12.6

$$I=\iint_D \frac{2e^{x^2}+3e^{y^2}}{e^{x^2}+e^{y^2}}\,dxdy \qquad D=\{(x,y)\mid 0\le x\le 1,\ 0\le y\le 1\}$$

$$I=\frac{1}{2}\iint \frac{2e^{x^2}+2e^{y^2}+3e^{y^2}+3e^{x^2}}{e^{x^2}+e^{y^2}}\,d\sigma=\frac{1}{2}\iint \frac{5e^{x^2}+5e^{y^2}}{e^{x^2}+e^{y^2}}\,d\sigma$$

$$=\frac{5}{2}\iint d\sigma=\frac{5}{2}$$

习12.7

设区域 $D=\{(x,y)\mid x^2+y^2\le 1,\ x\ge 0\}$

计算 $I=\iint_D \frac{1+xy}{1+x^2+y^2}\,dxdy$

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^1 \frac{1+r^2\cos\theta\sin\theta}{1+r^2}\,rdr$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\left[\int_0^1\frac{rdr}{1+r^2}+(\cos\theta\sin\theta)\cdot\int_0^1\frac{r^3dr}{1+r^2}\right]$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\left[\frac{1}{2}\ln(1+r^2)\Big|_0^1+\frac{(\cos\theta\sin\theta)}{2}\cdot\int_0^1 1-\frac{1}{1+r^2}\,dr^2\right]$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\frac{1}{2}\ln 2+\frac{\cos\theta\sin\theta}{2}\left[1-\int_0^1\frac{dr^2}{1+r^2}\right]\right]d\theta$$

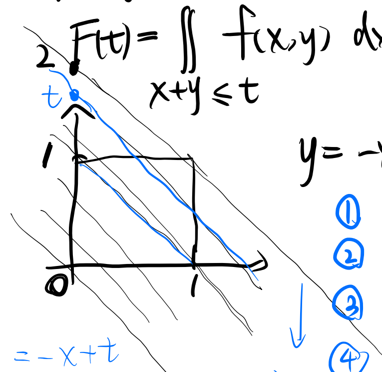
$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\frac{1}{2}\ln 2+\frac{\sin\theta\cos\theta}{2}(1-\ln 2)\right]d\theta$$

$$=\frac{1}{2}\pi\ln 2+\frac{1-\ln 2}{8}\int_0^{2\pi}\sin 2\theta\, d2\theta$$

$$=\frac{1}{2}\pi\ln 2$$

习12.8

$F(t)=\iint_{x+y\le t} f(x,y)\,dxdy$，其中 $f(x,y)=\begin{cases}1, & 0\le x\le 1, 0\le y\le 1.\\ 0, & 其他\end{cases}$ 求 $F(t)$ 表达式



$y=-x+t$

① $t\in[2,+\infty)$ $\quad F(t)=1$

② $t\in[1,2)$ $\quad F(t)=1-\frac{(2-t)^2}{2}$

③ $t\in[0,1)$ $\quad F(t)=\frac{t^2}{2}$

④ $t\in(-\infty,0)$ $\quad F(t)=0$

$1=-x+t$

$x=t-1$

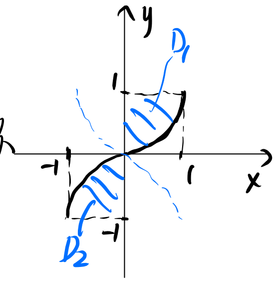
$1-(t-1)$

$=1-t+1$

300.12.1

D: $y=x^3$, $y=-1$, $y=1$ 及 y 轴

以D为底以 $z=x^3+y$ 为顶的曲顶柱体体积



$x^3+y=0 \Rightarrow y=-x^3$

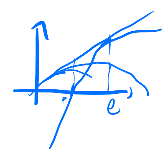
$\Rightarrow$ D_1 $z>0$ D_2 $z<0$

体积 $\iint\limits_{D_1} z\,d\sigma - \iint\limits_{D_2} z\,d\sigma$ (D)

300.12.2

$x=0$, $y=0$, $x+y=\frac{1}{4}$, $x+y=1$ 围成

若 $I_1=\iint\limits_D [\ln(x+y)]^3\,dxdy$



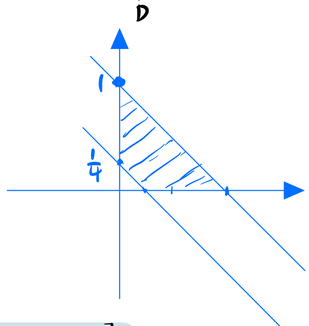
$I_2=\iint\limits_D (x+y)^3\,dxdy$

$I_3=\iint\limits_D [\sin(x+y)]^3\,dxdy$

$x\in(0,1)$: $\ln x<0$

$0<\sin x<x$

保号性 $\Rightarrow$ $I_1<I_3<I_2$



300.12.3

$I_1=\iint\limits_D (x^3+y^2)\,dxdy$ $D=\{(x,y)\mid x^2+y^2\le 4\}$

$I_2=\iint\limits_D (x^3-y^2)\,dxdy$

$I_3=\iint\limits_D (x^3+y^3)\,dxdy$

$I_1=\iint\limits_D y^2\,dxdy$

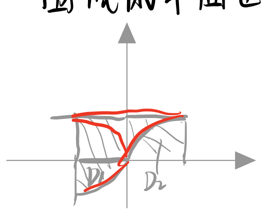
$I_2=\iint\limits_D -y^2\,dxdy$ $\Rightarrow$ $I_2<I_3<I_1$

$I_3=0$

300.12.4

设区域D是由曲线 $y=\sin x$ 与直线 $x=\pm\frac{\pi}{2}$, $y=1$

围成的平面区域，则 $\iint\limits_D (xy^5-1)\,d\sigma$



$\iint\limits_{D_1} xy^5-1\,d\sigma = \iint\limits_{D_2} xy^5-1\,d\sigma$

$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\sin^5 x-1\,dx = -\pi + 2\int_0^{\frac{\pi}{2}} x\sin^5 x\,dx$

$$\iint_D (xy^5-1)\,d\sigma = -\iint_D d\sigma + 0 = -\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} dx \int_{\sin x}^{1} dy = -\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1-\sin x)\,dx$$
$$= -x+\cos x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = -\pi$$

① $\int_0^{\pi/2} x\sin^5 x\,dx = \int_0^{\pi/2} x\sin^4 x\, d(-\cos x)$
$= -\cos x\cdot x\cdot\sin^4 x\Big|_0^{\pi/2} + \int_0^{\pi/2}\cos x\,(\sin^4 x + 4x\sin^3 x\cdot\cos x)\,dx$
$= \int_0^{\pi/2}\sin^4 x\, d\sin x + 4\int_0^{\pi/2} x\sin^3 x(1-\sin^2 x)\,dx$
$= \frac{\sin^5 x}{5}\Big|_0^{\pi/2} + 4\int_0^{\pi/2} x\sin^3 x\,dx - 4\int_0^{\pi/2} x\sin^5 x\,dx$
$\Rightarrow 5\int_0^{\pi/2} x\sin^5 x\,dx = \frac{1}{5} + 4\int_0^{\pi/2} x\sin^3 x\,dx$
$\Rightarrow \int_0^{\pi/2} x\sin^5 x\,dx = \frac{1}{25} + \frac{4}{5}\int_0^{\pi/2} x\sin^3 x\,dx$

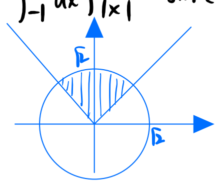
② $\int_0^{\pi/2} x\sin^3 x\,dx = \int_0^{\pi/2} x\sin^2 x\, d(-\cos x)$
$= -\cos x\cdot x\cdot\sin^2 x\Big|_0^{\pi/2} + \int_0^{\pi/2}\cos x\cdot(\sin^2 x + 2x\sin x\cos x)\,dx$
$= \int_0^{\pi/2}\sin^2 x\, d\sin x + 2\int_0^{\pi/2} x\sin x(1-\sin^2 x)\,dx$
$= \frac{\sin^3 x}{3}\Big|_0^{\pi/2} + 2\int_0^{\pi/2} x\sin x\,dx - 2\int_0^{\pi/2} x\sin^3 x\,dx$
$\Rightarrow 3\int_0^{\pi/2} x\sin^3 x\,dx = \frac{1}{3} + 2\int_0^{\pi/2} x\sin x\,dx$
$\Rightarrow \int_0^{\pi/2} x\sin^3 x\,dx = \frac{1}{9} + \frac{2}{3}\int_0^{\pi/2} x\sin x\,dx$

③ $\int_0^{\pi/2} x\sin x\,dx = \int_0^{\pi/2} x\,d(-\cos x) = -x\cos x\Big|_0^{\pi/2} + \int_0^{\pi/2}\cos x\,dx = \sin x\Big|_0^{\frac{\pi}{2}} = 1$

$\Rightarrow I = -\pi + 2\left\{\frac{1}{25} + 4\left[\frac{1}{3} + 2\times 1\right]\right\}$
$= -\pi + 2\left[\frac{1}{25} + \frac{4}{3} + 8\right]$
$= -\pi + \frac{1406}{75}$

300.12.5

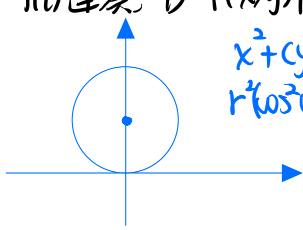
$\int_{-1}^{1} dx \int_{|x|}^{\sqrt{2-x^2}} \sin(x^2+y^2)\,dy$



$I = \int_{\pi/4}^{3\pi/4} d\theta \int_0^{\sqrt{2}} \sin r^2\, r\,dr$
$= \frac{1}{2}\int_{\pi/4}^{3\pi/4} d\theta \int_0^{\sqrt{2}} \sin r^2\, dr^2$
$= \frac{1}{2}\int_{\pi/4}^{3\pi/4} 1-\cos 2\, d\theta$
$= \frac{\pi}{4}(1-\cos 2)$ (B)

300.12.6

$f(t)$连续, $D=\{(x,y)\mid x^2+y^2\leq 2y\}$, $\iint_D f(xy)\,dxdy = ?$



$x^2+(y-1)^2=1$
$r^2(\cos^2\theta+\sin^2\theta)\leq 2r\sin\theta$
$r\leq 2\sin\theta$

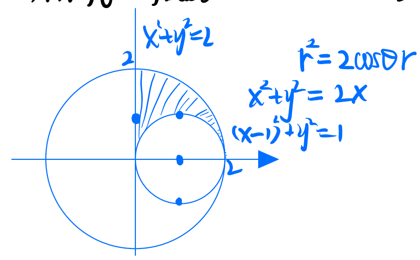
$\int_0^{\pi} d\theta \int_0^{2\sin\theta} f\cdot r\,dr$ (D)

300. 12.7

设 $f(u)$ 为连续函数，则二次积分 $\int_0^{\frac{\pi}{2}} d\theta \int_{2\cos\theta}^{2} f(r^2)\, r dr$ 在直角标系下化二次积分

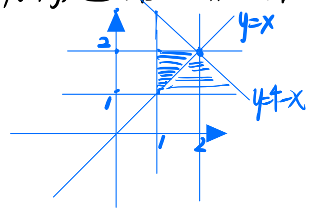
$$\int_0^{\frac{\pi}{2}} d\theta \int_{2\cos\theta}^{2} f(r^2)\, r dr$$

$$= \int_0^2 dx \int_{\sqrt{1-(x-1)^2}}^{\sqrt{4-x^2}} f(x^2+y^2)\, dy$$



300. 12.8

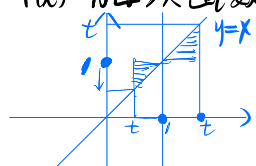
$f(x,y)$ 连续，$\int_1^2 dy \int_1^y f(x,y)\, dx + \int_1^2 dy \int_y^{4-y} f(x,y)\, dx$



$$\int_1^2 dy \int_1^{4-y} f(x,y)\, dx$$

300. 12.9

设 $f(x)$ 为连续函数，$F(t)=\int_1^t dy \int_y^t f(x)dx$，则 $F'(t)=?$



$t\geq 1$ 时 $F(t)=\frac{1}{2}(t-1)^2$

$t<1$ 时 $F(t)=\frac{1}{2}(1-t)^2$

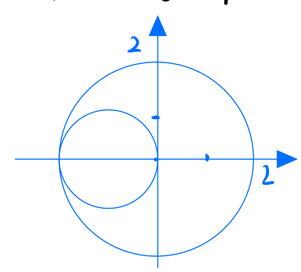
$F(t)=\frac{1}{2}(t-1)^2$

$F'(t)=(t-1)\, f'(t)$

$F(t)=\int_t^1 dy \int_t^y f(x)dx$

300. 12.10

求 $\iint_D (\sqrt{x^2+y^2}+y)\, d\sigma$，其中 D 是由圆 $x^2+y^2=4$ 和 $(x+1)^2+y^2=1$ 所围成的平面区域，如图所示



$r^2\cos^2\theta + 1 + 2r\cos\theta + r^2\sin^2\theta = 1$

$r + 2\cos\theta = 0$

$r = -2\cos\theta$

$$2\iint_D r + r\sin\theta \, d\sigma$$

$$= 2\int_0^{\pi} d\theta \left[\int_0^2 r(1+\sin\theta)\, r dr + \int_0^{2\cos\theta} r(1+\sin\theta)\, r dr\right]$$

$$= \frac{16}{3}\int_0^{\pi} (1+\sin\theta)(8+8\cos^3\theta)\, d\theta$$

$$= \frac{16}{3}\left[\pi - 2\right]$$